Nakyoung AI

Self-Hosted AI Chat ServerSystem

2026

In Progress

End-to-End Devloper

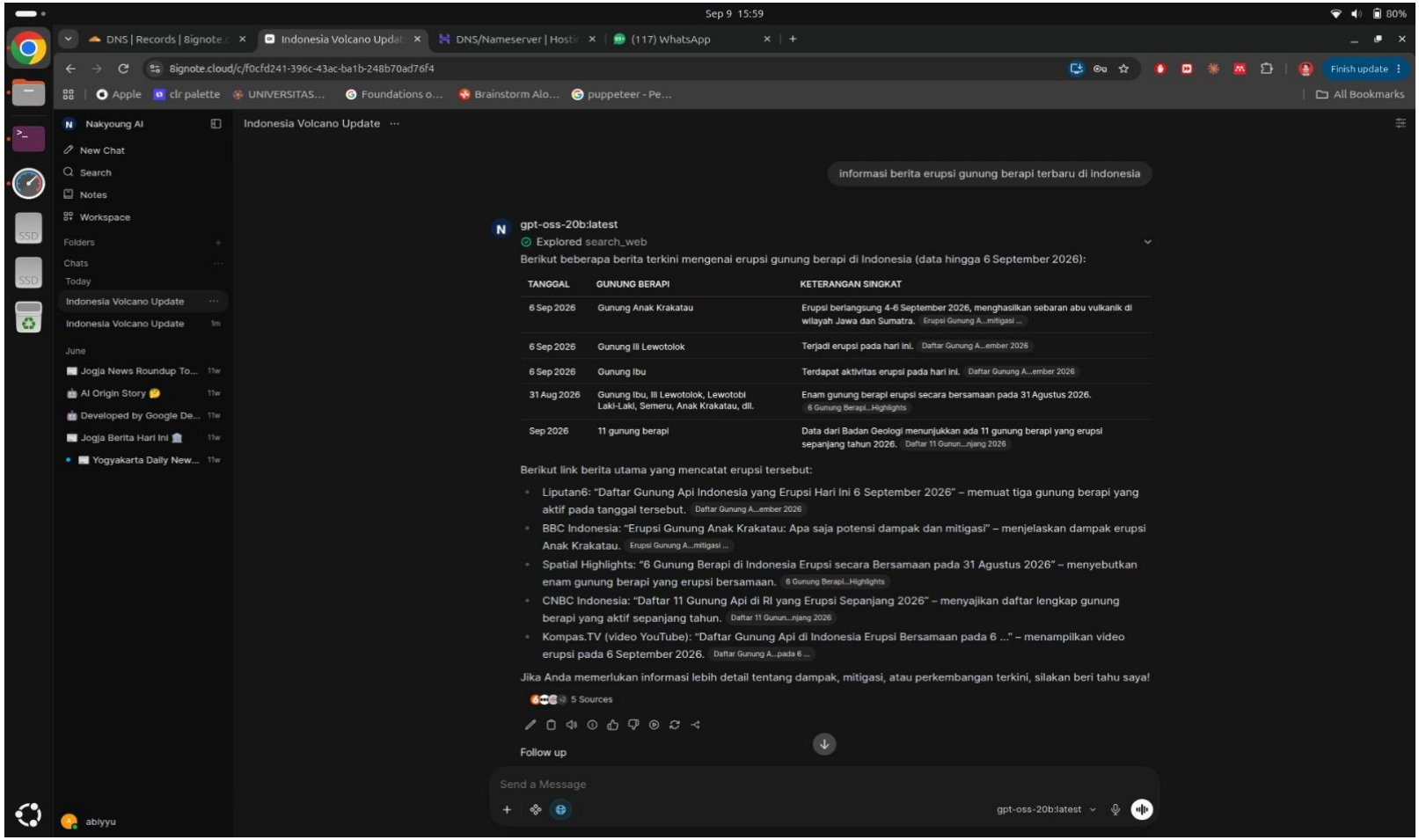
System Design, database, deployment — Self-initiated freelance project.

Abiyyurahman.vercel.app/gallery

8ignote.cloud

Description

I wanted to run my own AI chat service on hardware I actually own, not a wrapper around someone else's API. I turned my personal laptop into a server running Gemma 4 and gpt-oss-20b through Ollama, GPU-accelerated on an RTX 4050, exposed publicly through Cloudflare Tunnel on a custom domain (8ignote.cloud) without opening a single port, with a custom-branded chat interface and real-time web search through a self-hosted SearXNG instance, no paid API key involved. The service runs in production, though uptime depends on my personal laptop being powered on, this isn't a 24/7 hosted service, so the link may occasionally be unreachable. Keeping it reliable meant real debugging rather than accepting something that technically worked but quietly failed: a software bug that kept crashing one of the models, a networking misconfiguration that broke public access, and an issue where the AI would confidently make up sources whenever its web search silently failed behind the scenes. The part that stuck with me most was that last issue, the model wasn't malfunctioning, it was doing exactly what AI does without real information: producing a plausible-sounding answer anyway. That changed how I think about reliability, a system that fails loudly is easy to fix, one that fails silently while still sounding confident is far more dangerous.



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Portfolio Submission for

Apple Developer Academy Indonesia